Calibrating Temperature Sensors

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Introduction and Setup

In this experiment, we calibrated an NTC thermistor using a **DS18B20** digital thermometer as the reference. The DS18B20 works from −55 to +125 °C, is accurate to ±0.5 °C, and has a 9–12 bit resolution.

The circuit is a simple voltage divider with the NTC connected on one side to GND. (We tried both configurations and got approximately similar readings within a 99% confidence interval.) A 10-bit ADC (0–1023 counts) reads the divider. For the calibration we collected 3 runs: one heating and two cooling. After cleaning, we had 1078 heating and 4243 cooling samples from 0 to 98 °C. We fitted the readings in two ways: polynomials of degree 1 to 3, and the physical NTC β-model. We also measured how fast the thermistor responds (its time constant).

Raw Data

Temperature falls steadily as ADC counts rise, but not linearly: the curve is steeper at low counts. The time plots show a slow, smooth change in bath temperature.

Heating and Cooling Calibration

Table 1. Data audit before fitting.

Item	Heating	Cooling
Samples logged	1079	4252
Empty (NaN) rows	1 (ADC 849)	2
Outliers removed ($> 3\sigma$)	0	7 (> 1.66 °C)
Samples used in fit	1078	4243

We fitted polynomials of degree 1 to 3 to temperature against ADC. Empty rows and outliers (beyond 3 standard deviations of the first cubic fit, or 1.66 °C for the cooling runs) were removed first. We used the common 3σ cut; the earlier 2.5σ cut removed 2 more points and changed the RMSE by less than 0.01 °C.

Table 2. Fit errors (°C) for heating (H, $N = 1078$) and cooling (C, $N = 4243$).

Deg.	R^2 (H)	RMSE (H)	Max (H)	R^2 (C)	RMSE (C)	Max (C)
1	0.99600	2.326	3.913	0.99265	1.462	4.010
2	0.99718	1.951	4.146	0.99882	0.586	2.752
3	**0.99991**	**0.345**	**1.459**	**0.99898**	**0.546**	**1.727**

The cubic fit is the best. The straight line leaves an S-shaped error of up to about 4 °C. The cubic cuts the RMSE to 0.35 °C for heating and 0.55 °C for cooling. All cubic terms are significant ($|t| > 18$, $p < 0.001$). The large condition number ($\sim 10^9$) comes from using raw ADC powers and is harmless. The cooling errors are correlated in time (Durbin–Watson 0.02), so the standard errors are slightly too small.

Table 3. Cubic fit coefficients. All terms have $p < 0.001$.

Coefficient	Heating	Cooling	Global	Global std. err.
c_0 (intercept)	−18.792	−8.433	−17.559	0.076
c_1 (ADC)	0.2277	0.1562	0.2134	0.001
c_2 (ADC²)	-3×10^{-4}	-1×10^{-4}	-2.495×10^{-4}	1.34×10^{-6}
c_3 (ADC³)	2.107×10^{-7}	1.186×10^{-7}	1.869×10^{-7}	9.36×10^{-10}
N / AIC	1078 / 773.8	4250 / 7028	5321 / 8557	—

Table 4. Residual checks for the cubic fits.

Statistic	Heating	Cooling	Global
F-statistic	4.06×10^6	1.35×10^6	3.21×10^6
Durbin–Watson	0.156	0.022	0.031
Jarque–Bera	7.80	321.1	468.2
Skew / kurtosis	0.21 / 2.98	−0.62 / 2.48	−0.72 / 2.80
Condition number	2.6×10^9	1.3×10^{10}	2.4×10^9

The heating residuals are close to normal (skew 0.2, kurtosis 3). The cooling and global residuals are slightly skewed, which means a small shape error remains, so the 95 % band is a rough guide.

Heating vs. Cooling Hysteresis

We compared the two cubic fits over the shared range (ADC 309–849). The biggest gap is **1.15 °C** (1.15 % of full scale) and the average gap is **0.60 °C** (Figure 1). There are three likely causes:

1. **Thermal lag.** The NTC and the reference probe have different time constants $\tau = R_{\text{th}}C_{\text{th}}$, so one sensor trails the other while the temperature changes.
2. **Maths effect.** ΔT is the difference of two cubics, so it is a smooth wave crossing zero (near ADC ≈ 500).
3. **Water bath.** Without good stirring, the two probes see different temperatures.

The average gap is smaller than the reference uncertainty, so we do not claim real sensor hysteresis.

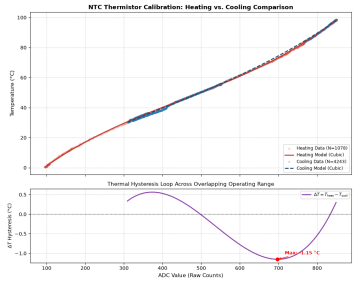


Figure 1. Heating vs. cooling fits and the gap ΔT.

Global Calibration and 95 % Prediction Band

We joined the heating data and the cleaned cooling data ($N = 5321$) and fitted one cubic, where a is the ADC count:

$$T = 1.869 \times 10^{-7}a^3 - 2.495 \times 10^{-4}a^2 + 0.2134a - 17.559 \quad (1)$$

It has $R^2 = 0.999$ and RMSE 0.54 °C. The mean 95 % prediction half-width is **±1.06 °C** (Figure 2).

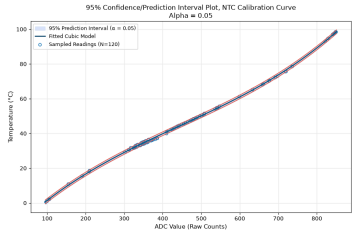


Figure 2. Global cubic fit with its 95 % prediction band.

Theoretical NTC β-Model

The divider output is $V_{\text{out}} = V_{\text{cc}}R_{\text{f}}/(R_{\text{NTC}} + R_{\text{f}})$. The ADC uses V_{cc} as its reference, so the reading does not change if the supply drifts:

$$\mathrm{ADC} = \mathrm{ADC}_{\max} \cdot \frac{R_f}{R_{\mathrm{NTC}} + R_f} \quad \Rightarrow \quad r \equiv \frac{R_{\mathrm{NTC}}}{R_f} = \frac{1023 - \mathrm{ADC}}{\mathrm{ADC}} \tag{2}$$

In an NTC, heat lifts charge carriers across the band gap E_g. This gives the Arrhenius law $R(T) = R_0 \exp[\beta(1/T - 1/T_0)]$, with $\beta = E_g/2k_B$ and $T_0 = 298.15$ K. Dividing by R_f and taking the log gives a straight line:

$$\ln r = \frac{\beta}{T} + c, \quad c = \ln r_0 - \frac{\beta}{T_0}. \tag{3}$$

Let $A = 1/\beta$ and $B = -c/\beta$. Then solving for temperature gives the formula used in the logger:

$$T = \frac{1}{A \ln((1023 - \mathrm{ADC})/\ \mathrm{ADC}) + B} - 273.15. \tag{4}$$

β-Model Validation

We fitted a straight line to $\ln r$ against $1/T$ (Figure 3). It gives $\beta = 3965.3$ K, $R^2 = 0.9993$ and an RMSE of **0.598 °C**, using only two parameters (β and r_0).

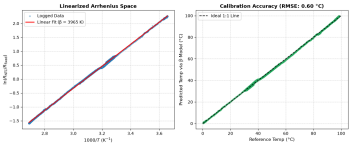


Figure 3. Straight-line Arrhenius fit (left) and β-model accuracy (right).

Response Time of the Calibrated NTC

We dipped the probe repeatedly into ice water (0 °C) and room-temperature water, giving a square-wave input. After removing one spike (at 496 s) and the first cycle, 3 full periods remained. We converted ADC to °C with both the cubic and the β-model, then fitted Newton's law with a start delay to the 3 periods together:

$$T(t) = T_f + (T_i - T_f)e^{-(t-t_0)/\tau}. \tag{5}$$

The 3 periods share τ, T_i and T_f; each has its own delay t_0. The error comes from leaving one period out at a time (Figure 4).

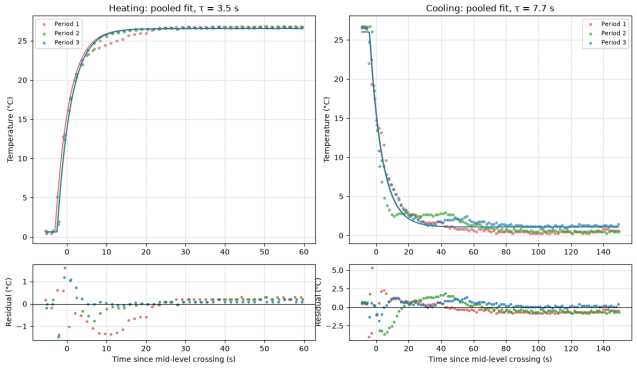


Figure 4. Pooled fit (cubic calibration) of the 3 heating and 3 cooling steps.

Table 5. Response time from the pooled fits (± is the leave-one-out error).

Step	Calibration	τ (s)	10–90 % (s)	RMSE (°C)
Heating	Cubic	3.5 ± 0.3	7.8	0.47
Heating	β-model	3.3 ± 0.3	7.3	0.50
Cooling	Cubic	7.7 ± 2.2	16.8	0.92
Cooling	β-model	8.6 ± 2.2	18.9	1.02

The thermistor reaches 63 % of a step in about **3.4 s** when heating and about **8 s** when cooling, so it needs roughly 17 s and 40 s to settle (5 τ). The two calibrations agree within 1 s, less than the error, so the choice does not matter. Cooling is slower and less certain: one period was much slower, and the ice bath warmed by about 2 °C during period 2, which a single exponential cannot follow (RMSE about 1 °C).

Sensitivity and Error Theory

The sensitivity $dT/d\mathrm{ADC}$ is 0.102–0.195 °C/count for the cubic and 0.101–0.245 °C/count for the β-model (Figure 5). The sensor reads best mid-range and worse near both ends, where the divider output flattens.

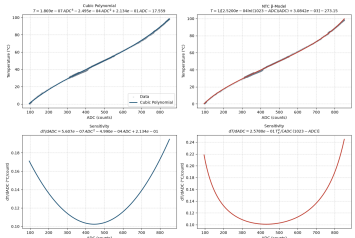


Figure 5. Fitted curves and sensitivity $dT/d\mathrm{ADC}$ for both models.

RMSE only measures scatter around the fit. To predict the error from its sources, we propagate three independent ones, $u_T^2 = \sum_i (\partial T/\partial x_i)^2 u_{x_i}^2$:

$$u_T = \sqrt{u_{\mathrm{ref}}^2 + (dT/d\mathrm{ADC})^2 u_{\mathrm{ADC}}^2 + \boldsymbol{g}^T \boldsymbol{C} \boldsymbol{g}}. \tag{6}$$

$u_{\mathrm{ref}} = 0.5$ °C is the DS18B20 accuracy. The ADC rounds to the nearest count ($u_{\mathrm{ADC}} = 0.5$), giving $u_{\mathrm{res}} = |dT/d\mathrm{ADC}| \cdot 0.5$, so a flat region reads coarsely. The fit term uses the parameter covariance $\boldsymbol{C}$ and $\boldsymbol{g} = \partial T/\partial$ params (with cross terms, as the coefficients are correlated).

Table 6. Error theory compared with the measured errors.

Quantity (°C)	Cubic	β-model
Resolution, $\lvert dT/d\mathrm{ADC}\rvert_{\max}$ (°C/count)	0.195	0.245
u_{res} (max)	0.097	0.122
u_{fit} (mean / max)	0.016 / 0.037	0.012 / 0.026
u_{ref}	0.5	0.5
u_T (mean / max)	**0.505 / 0.510**	**0.505 / 0.515**
Measured RMSE	0.54	0.60
Largest residual	1.56	1.46

The reference error dominates; the fit and resolution terms add less than 0.1 °C each. The theory gives $u_T \approx 0.51$ °C, close to the measured RMSE (0.54 and 0.60 °C). The small extra part (~ 0.2 –0.3 °C) is what the theory leaves out: thermal lag, hysteresis and model shape error. We quote the **largest** u_T over the range as the standard error: **±0.51 °C (cubic)** and **±0.52 °C (β-model)**, or about ±1.0 °C at 95 % (two standard deviations), which agrees with the ±1.06 °C prediction band.

Conclusion

Both models calibrate the thermistor to within the ±1 °C uncertainty of the DS18B20. The global cubic (4 parameters) has RMSE 0.54 °C and ±1.06 °C (95 % PI); the β-model (2 parameters, $\beta = 3965$ K, $R^2 = 0.9993$) has RMSE 0.60 °C. Error theory gives the same ±0.5 °C for both because the reference dominates. The cubic is slightly better inside 0–98 °C. The β-model is almost as good with half the parameters, works well outside the range, and gives a real physical constant.

Limits. The DS18B20 is only accurate to ±0.5 °C, so smaller errors cannot be proved. Lag between the two sensors adds error during fast changes. Only three runs were usable (0–98 °C), so do not use the cubic outside this range.

Recommendation. Use the β-model as the main calibration, and the cubic only when you need the lowest error inside 0–98 °C. The heating/cooling gap (about 0.6 °C) is below the reference uncertainty, so one calibration works for both directions.